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Project Title:	Skeletal Muscle Growth and Repair During Microgravity		
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Task Description:	As the duration of planned space missions increases, so does the probability of increased levels of muscle atrophy and potential for muscle injury. In addition, little is understood about the impact of weightlessness on muscle growth. This study uses a ground-based model of hindlimb suspension to examine the effects of weightlessness on the efficiency of skeletal muscle growth and regeneration. The aim is to determine the cellular changes that occur in skeletal muscle fibers and their satellite cells during weightlessness and understand how these changes alter the growth potential and potential for recovery after injury.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	We have learned that skeletal muscles in the unweighted environment do not grow and, if injured, do not properly regenerate to repair the injury. The growth and repair process is not completed properly even if the muscles are returned to normal load environment. These studies suggest that growing or immature skeletal muscles or regenerating muscles must be subjected to some load during the developmental process in order for full potential to be realized.
Task Progress:	Summary Satellite cells have an integral role in skeletal muscle growth and regeneration. In immature animals the cells are a dividing population and the progeny function to provide additional myonuclei to enlarging fibers. Since muscles within the body grow at different rates, the mitotic and fusion behavior of satellite cells in each muscle is regulated in a way that reflects the growth characteristics of each individual muscle. Unweighting the growing muscle, as in microgravity, completely disrupts the growth process and the hallmark of that disruption is the cessation of satellite cell mitotic and fusion behavior. Finally, when weightbearing is reinstated at normal gravity, the process of myofiber growth including satellite cell mitosis and fusion is reinstated, but the growth deficit incurred during the period of weightlessness remains. That is, there is no compensation for the period of reduced growth and the muscles retain a permanent growth deficit. We hypothesized that weightlessness somehow alters the satellite cell population rendering it unable to respond to the demands of growth following a period at 0 gravity. The behavior of the cells is regulated in part by the expression of muscle-specific helix-loop-helix family of transcription factors (MRF's) that are differentially expressed during the functional life history of the cells. For example, MyoD and Myf 5 expression is associated with actively dividing cells, whereas myogenin expression is associated with cells entering terminal differentiation. Pax 7 is a factor thought to be involved with the specification of mesenchymal cells to the myogenic lineage during embryonic development. For unknown reasons, Pax expression persists in satellite cells during postnatal development. Our approach was to characterize the satellite cell population during normal growth by demonstrating how expression of these factors was compartmentalized. By characterizing the expression profile of the cells the goal was to demonstrate if there was a loss of a particular compartment of the cells during weightlessness. The working hypothesis was that a compartment of fusion competent, myogenin positive cells required for normal growth was depleted or lost during the unweighting period. The first step to test this hypothesis was to investigate and quantify the component parts of the satellite cell population by studying the MRF expression profile. The expression profile of satellite cells in growing muscles was determined by immunostaining soleus and EDL muscles of immature rats for the presence of MyoD, myogenin and Pax 7. We first showed that staining for these three factors marked all of the satellite cells present in both soleus and EDL muscles. The majority of satellite cells express Pax while MyoD and myogenin both occurred with about half the frequency of Pax. However, we also found that the staining procedure overestimated the satellite cell population in each muscle. We reasoned that the overestimation was because cells co-expressed the factors and/or cells stained with the markers that were outside of the conventional satellite cell position. We subsequently found both possibilities to be true. In both the soleus and EDL muscles the pattern of staining was similar and labeling was found outside the conventional satellite cell position and the cells in each muscle co-expressed the markers. Sections were co-stained for dystrophin or laminin and each of the MRF's in order to determine the position of labeled nuclei either within myofibers (myonuclei) or within the interstitial compartmtent. Perhaps the most surprising result of the study was that a percentage o myonuclei in growing muscles stained for MyoD, myogenin or Pax. The incidence of MRF stained myonuclei was higher in the EDL than in the soleus. While MyoD (4%) and myogenin (2%) stained myonuclei could be the result of transient expression following the fusion of a satellite cell with the fiber, Pax (6%) expression could only be the result of post-fusion upregulation. These results suggest that Pax, MyoD and myogenin have possible functions in myofibers of growing fibers. The incidence of staining in the interstitial compartment was about the same in both muscles. Pax (13%), MyoD (1%) and myogenin (5%) were randomly distributed throughout the interstitial compartment in each muscle. These findings are similar to other studies documenting myogenic cells in the interstitial space and suggest that satellite cells are capable of migrating between myofibers in postnatal muscle and that myogenic cells arising from other sources may enter (and leave) the satellite cell compartment. Taking into account the cells outside the satellite cell compartment we next determined how expression of the three markers was distributed within the population including the amount of co-expression within the satellite cell population. The relative expression pattern within the satellite cell populations of the two muscles was very similar. In the soleus muscle 43% of the satellite cell population expressed Pax-only, 8% MyoD-only and 20% myogenin-only, while .6% expressed Pax/myogenin, 18% Pax/MyOD and 8% MyoD/myogenin. In the EDL muscle there was 31% Pax-only, 7% Myod-only, 17% myogenin-0nly, 2% Pax/myogenin, 37% Pax/MyoD and 10% MyoD/myogenin. Thus the major difference in the two muscles was largely in the MyoD expression compartment and, in turn the Pax and Pax/MyoD expression compartments. We next compared the expansion compartments within the satellite cell population by injecting BrdU to label dividing cells. One hour following injection muscles were harvested and co-stained for BrdU and each of the MRF's to determine where satellite cells would be located within the expression profile constructed. In both muscles, the majority of BrdU was associated with the MyoD positive cells. Pax7/BrdU and Myogenin/ BrdU positive cells were more numerous in the soleus compared to the EDL. The distribution of BrdU label was fitted within the Pax7, MyoD, and myogenin co-expression profile to identify the expansion compartments within the satellite cell population for each muscle (Figure 1). In both muscles, label was found exclusively within the MyoD compartment. However, the majority of BrdU-labeled MyoD cells also co-expressed Pax. A small number of dividing cells co-expressed myogenin. These results show that satellite cells are partitioned into functional compartments during the time they are actively producing myonuclei during postnatal growth. Taken together with previous studies these data suggest that the Pax compartment is a slowly dividing population of cells and represent the reserve population of satellite cells where growth potential is stored. Hence one daughter cell of a Pax-only mitosis remains within the Pax reserve compartment. The other daughter cell co-expresses MyoD and enters the expansion compartment where it divides multiple times. The significance of the continued expression of Pax may be that the cell can re-enter the mitotically quiescent reserve compartment if conditions dictate. After several divisions Pax expression is downregulated and the cells enter the MyoD compartment for additional divisions. The cells become committed to terminal differentiation with upregulation of myogenin. In this manner a constant supply of myonuclei is produced and the population of satellite cells continues to maintain a reserve population as well as a population of fusion competent cells. In addition, cells within the dividing compartment retain the capability of becoming quiescent reserve cells that can be quickly recruited to produce myonuclei under conditions that promote hypertrophy or require repair. While we were unable to complete our studies by actually studying the changes in the expression profile of satellite cells during and after weightlessness our original hypothesis can now be tested in the future. Our previous studies showed that mitotic activity of satellite cells stops with the onset of the weightless environment. However, muscle growth and addition of myonuclei is not as sensitive and continues for a period, after which it also slows and stops. Because of the lag in cessation of muscle growth, fusion competent myogenin cells continue to fuse with their myofibers despite the fact that no more are being produced. Consequently, the myogenin compartment becomes depleted. When the animals are returned to a weightbearing, their muscles fail to go through any compensatory growth period to make up for the duration during which growth was not

	taking place because the population that was lost must be first reconstituted before accretion of nuclei can occur. The results suggest a feedback mechanism between compartments has been disrupted. The presence of a substantial population of myogenic cells in the interstitial space was surprising. We had conducted experiments previously that suggested that muscle regeneration was dependent solely upon the satellite cell population in the muscle at the time of injury. We repeated these experiments using LacZ mice. All cells in the mice expressed LacZ which could be visualized using either antibody or cytochemistry. By placing non-LacZ muscles in the LacZ positive recipients we again found that the regeneration that takes place in this free graft model is dependent only upon the myogenic cells within the muscle. There is little invasion of host myogenic cells into the graft. Thus, despite the suggestion that myogenic cells are freely moving within the interstitial space and may be derived from non-satellite cell sources, they do not appear to make any significant contribution to regeneration.
Bibliography Type:	Description: (Last Updated: 04/02/2007)
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